

Blockchain and Application Stack

L3 Data

PostgreSQL
16 (20
entities, 3NF)
Redis 7 · IPFS

L4 API and Services

Express /
Node 20 ·
FastAPI
WebSocket ·
EIP-712 ·
MCP
middleware

L5 Presentation

React 18 +
Vite
ERC-4337 AA
· MetaMask /
email



L1 Settlement / Network

Polygon
zkEVM
Cardona (ZK
proofs)
Ethereum
Sepolia ·
CCIP · The
Graph ·
Tenderly

L2 Smart-Contract Platform

Solidity 0.8.20
·
OpenZeppelin
v5
UUPS ·
TimeLock ·
RBAC